SUMMER INSTITUTIONAL TRAINING REPORT

Duration:

(from 1th June, 2026 to 15th July 2026)

SUBMITTED BY:

KARAN SAH

5th Sem. Section B

Class Roll No.: 207/24

University Roll No.: 2427541

Department of Computer Science & Engineering DAV Institute of Engineering & Technology Jalandhar, India

(2024-2028)

WATCH CRAFT

A Movie Analytics and Visualization Platform

Final Project Report

Submitted By
Karan Sah (207/24)

B.Tech Computer Science and Engineering

Submitted To
Mr. Harkirat Singh

D.A.V Institute of Engineering and Technology

6 Week Summer Training Program

Industrial Training Partner: O7 Solutions, Jalandhar

Certificate

This is to certify that the project titled 'WATCH CRAFT: A Movie Analytics and Visualization Platform' submitted by Karan Sah (207/24) during the 6 week Summer Training Program is a bonafide work carried out under the guidance of Mr. Harkirat Singh and in collaboration with O7 Solutions.

Declaration

I hereby declare that this report is my original work and has not been submitted to any other institution or university for any degree or diploma.

Acknowledgement

I express my sincere gratitude to Mr. Harkirat Singh for his guidance and support throughout the development of Watch Craft. I also thank O7 Solutions for providing industrial exposure and mentorship during the summer training program.

Abstract

Watch Craft is a modern movie analytics platform developed using Python and Streamlit. It combines interactive visualizations, multi-platform rating analysis, genre intelligence, streaming platform analytics and hidden gem discovery to provide users with a unique way to explore cinema through data science.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The entertainment industry generates massive amounts of data every day in the form of movie ratings, audience reviews, critic opinions, streaming statistics, and viewer preferences. While this information is widely available across multiple platforms, extracting meaningful insights from it remains difficult due to data fragmentation and the lack of integrated analytical tools.

Data Science and Data Visualization provide powerful techniques for transforming raw information into actionable insights. Interactive dashboards allow users to identify trends, compare performances, and discover hidden patterns that would otherwise remain unnoticed.

Watch Craft was developed to address this challenge by creating an interactive movie analytics platform capable of converting movie datasets into meaningful visual stories.

1.2 Introduction to Movie Analytics

Movie analytics refers to the process of collecting, processing, and analyzing movie-related data to generate insights regarding audience behavior, genre popularity, rating trends, and content distribution.

Modern movie analytics can help answer questions such as:

- Which genres perform best among audiences?
- How do critics and audiences differ in their opinions?
- Which streaming platform offers the strongest content library?
- Which highly rated movies remain underrated or undiscovered?

The answers to these questions can be effectively communicated using visual analytics.

1.3 Need for Movie Analytics Platforms

Most existing movie websites focus primarily on displaying information rather than analyzing it.

Current platforms often suffer from:

- Fragmented rating systems.
- Limited analytical capabilities.
- Lack of visual insights.
- Difficulty in discovering underrated movies.
- Poor comparison between review platforms.

An integrated analytical platform can significantly improve movie discovery and user understanding.

1.4 Problem Statement

Movie enthusiasts frequently consult multiple websites such as IMDb, Rotten Tomatoes, Letterboxd, and streaming platforms before deciding what to watch.

This process creates several problems:

- Information is scattered across platforms.
- Ratings often conflict with each other.
- Hidden gems remain difficult to identify.

- Genre trends are not easily visible.
- Streaming availability is difficult to compare.

Therefore, there exists a need for a centralized movie analytics platform capable of integrating these insights into a single environment.

1.5 Proposed Solution

Watch Craft is proposed as an interactive movie analytics and visualization platform developed using Python and Streamlit.

The platform integrates:

- Multi-platform ratings
- Genre analysis
- Streaming analytics
- Hidden gem detection
- Recommendation support
- Interactive dashboards

The objective is to transform movie exploration into a visual and analytical experience rather than a simple search process.

1.6 Objectives of the Project

The primary objectives of Watch Craft are:

- To develop an interactive movie analytics dashboard.
- To compare ratings from multiple platforms.
- To analyze audience and critic behavior.

- To study genre popularity and trends.
- To provide streaming platform insights.
- To identify hidden gems within the dataset.
- To improve movie discovery using data science techniques.

1.7 Scope of the Project

The scope of Watch Craft includes:

- Movie dataset exploration.
- Interactive visualizations.
- Statistical analysis.
- Genre intelligence.
- Streaming analytics.
- Hidden gem discovery.

The current implementation uses a curated movie dataset and operates as a web-based dashboard deployed using Streamlit.

1.8 Applications

The platform can be useful for:

Movie Enthusiasts

To discover movies and explore trends.

Researchers

To study audience behavior and entertainment patterns.

Students

To understand practical applications of data science.

Streaming Platforms

To analyze content distribution and genre preferences.

1.9 Project Overview

Watch Craft was developed using modern data science technologies including Python, Streamlit, Pandas, and Plotly.

The platform provides:

- Interactive dashboards
- Dynamic filtering
- Visual storytelling
- Real-time user interaction

The system combines analytical capabilities with an engaging cinematic user experience.

1.10 Chapter Summary

This chapter introduced the motivation, objectives, and scope of Watch Craft. The project aims to bridge the gap between raw movie data and meaningful insights by leveraging data science and visualization techniques.

The following chapter presents the literature review and discusses existing work related to movie analytics, recommendation systems, and entertainment visualization.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The rapid growth of online streaming platforms and movie review websites has resulted in an enormous amount of movie-related data being generated every day. This data includes audience ratings, critic reviews, genre information, streaming availability, and user preferences.

Traditional movie websites primarily focus on information retrieval rather than analytical exploration. As a result, users often struggle to identify patterns, compare ratings across platforms, or discover hidden cinematic gems.

Recent developments in Data Science, Data Visualization, and Recommendation Systems have made it possible to transform raw entertainment data into meaningful insights and interactive experiences.

This chapter reviews existing work related to movie analytics, recommendation systems, visualization dashboards, and entertainment intelligence systems.

2.2 Evolution of Movie Information Systems

Early movie information systems primarily served as digital catalogs that stored movie titles, cast information, and release dates.

As internet technologies evolved, platforms such as IMDb and Rotten Tomatoes introduced:

- User ratings
- Critic reviews
- Audience reviews
- Popularity rankings

- Recommendation features

These systems significantly improved movie discovery but still relied heavily on static information presentation.

Modern users increasingly demand:

- Personalized recommendations
- Visual insights
- Cross-platform comparisons
- Interactive exploration

This demand has driven the evolution of movie analytics systems.

2.3 Movie Recommendation Systems

Movie recommendation systems are among the most widely studied applications of Machine Learning and Data Science.

The primary goal of recommendation systems is to suggest movies that align with user preferences and interests.

The three major approaches are:

Content-Based Filtering

Content-based filtering recommends movies based on similarities in:

- Genre
- Director
- Cast
- Language
- Story themes

For example, users who enjoy science fiction movies may receive recommendations for other science fiction titles.

Collaborative Filtering

Collaborative filtering uses historical user behavior to identify similar users and recommend movies based on collective preferences.

This approach powers recommendation systems used by:

- Netflix
- Amazon Prime Video
- YouTube

Hybrid Recommendation Systems

Hybrid recommendation systems combine content-based and collaborative filtering approaches to improve accuracy and overcome individual limitations.

Most modern streaming platforms rely on hybrid models.

Watch Craft currently uses filtering and dataset exploration techniques and provides a strong foundation for future AI-powered recommendation systems.

2.4 Role of Data Visualization in Analytics

Data visualization transforms complex numerical information into graphical representations that are easier to understand and interpret.

In movie analytics, visualizations help reveal:

- Rating trends
- Audience behavior
- Genre popularity
- Streaming distributions
- Correlations among review platforms

Studies have shown that visual dashboards improve analytical understanding significantly compared to traditional tables and spreadsheets.

Watch Craft heavily utilizes interactive visualizations to improve user engagement and insight generation.

2.5 Multi-Platform Rating Systems

Different movie platforms evaluate films using different methodologies.

IMDb

IMDb primarily reflects opinions of general audiences worldwide and uses a 10-point rating scale.

Rotten Tomatoes

Rotten Tomatoes separates reviews into:

- Critic Scores
- Audience Scores

This distinction helps users compare professional and public opinions.

Letterboxd

Letterboxd ratings are heavily influenced by movie enthusiasts and cinephiles.

Metacritic

Metacritic aggregates weighted critic reviews from professional publications.

The presence of multiple rating systems provides richer analytical opportunities but also introduces inconsistencies that require comparative analysis.

One of the major objectives of Watch Craft is to integrate these fragmented rating systems into a single analytical environment.

2.6 Audience-Critic Disagreement

Numerous studies have shown that audiences and professional critics frequently disagree in their evaluation of movies.

Critics often prioritize:

- Cinematography
- Screenwriting
- Direction
- Artistic value

Audiences generally emphasize:

- Entertainment value
- Emotional impact
- Action sequences
- Re-watchability

These differences create opportunities for identifying:

- Cult classics
- Polarizing films
- Hidden gems

The Audience-Critic comparison module implemented in Watch Craft is directly inspired by these findings.

2.7 Entertainment Dashboards

Interactive dashboards have become increasingly important in business intelligence and data science applications.

Modern dashboard systems provide:

- Real-time filtering
- Interactive charts
- Dynamic exploration
- Statistical summaries

Popular dashboard technologies include:

- Tableau
- Power BI
- Streamlit
- Dash

Watch Craft adopts Streamlit because of its simplicity, flexibility, and ability to integrate directly with Python data science libraries.

2.8 Hidden Gem Discovery

One major limitation of traditional ranking systems is their tendency to favor highly popular movies.

This creates popularity bias where excellent but lesser-known films receive limited visibility.

Recent research suggests that combining:

- Ratings
- Popularity metrics
- Critic scores
- Audience reactions

can help identify underrated masterpieces.

The Hidden Gem Detector implemented in Watch Craft follows this philosophy by selecting movies with strong ratings across multiple platforms while avoiding pure popularity-based recommendations.

2.9 Streaming Platform Analytics

The rise of streaming services has fundamentally transformed the movie industry.

Major platforms now include:

- Netflix
- Disney+
- Prime Video
- HBO Max

- Hulu
- Apple TV+

Each platform follows unique content acquisition and production strategies, leading to differences in genre distribution and audience demographics.

Watch Craft includes a dedicated Streaming Analytics module that compares content distribution and rating trends across multiple streaming platforms.

2.10 Existing Systems and Their Limitations

Although current movie platforms provide valuable information, several limitations remain.

Existing Systems	Limitations
IMDb	Limited visual analytics
Rotten Tomatoes	Focused primarily on reviews
Letterboxd	Limited statistical exploration
Streaming Platforms	Lack cross-platform comparison

None of these systems provide:

- Unified rating comparison
- Genre intelligence
- Hidden gem analysis
- Interactive dashboards
- Streaming analytics in one environment

This gap motivated the development of Watch Craft.

2.11 Research Gap

The literature review identified the following research gaps:

- Absence of integrated rating comparison systems.
- Lack of visual storytelling in movie analytics.
- Limited hidden gem discovery mechanisms.
- Insufficient streaming platform analysis.
- Fragmented entertainment datasets.

Watch Craft attempts to address these limitations through a unified movie analytics dashboard.

2.12 Summary

This chapter reviewed previous work related to movie recommendation systems, data visualization, entertainment dashboards, and streaming analytics.

The study revealed that existing platforms provide information but rarely provide meaningful analytical insights. The identified research gaps motivated the development of Watch Craft as an interactive movie analytics platform capable of combining visualization, multi-platform ratings, genre intelligence, and hidden gem discovery into a single system.

The next chapter presents the system analysis and discusses the problem statement, feasibility study, and proposed solution architecture for Watch Craft.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Introduction

System analysis is the process of studying an existing problem, identifying user requirements, and developing an efficient solution to overcome the limitations of current systems. It acts as a bridge between problem identification and system implementation.

In the context of movie analytics, users today have access to enormous amounts of information through websites such as IMDb, Rotten Tomatoes, Letterboxd, and streaming platforms. However, extracting meaningful insights from this information remains difficult due to fragmentation and lack of analytical tools.

Watch Craft was developed to address these issues by providing an integrated movie analytics and visualization platform capable of transforming raw movie data into meaningful visual insights.

3.2 Problem Statement

The modern entertainment industry produces thousands of movies and television shows every year. Although multiple platforms provide ratings and reviews, users still face several challenges while exploring movie information.

The major problems identified are:

- Ratings are distributed across multiple platforms.
- Users cannot easily compare audience and critic opinions.
- Existing systems provide limited analytical capabilities.
- Discovering underrated movies is difficult.
- Genre trends are not easily visible.

- Streaming platform comparisons are limited.

As a result, movie enthusiasts often visit multiple websites before making viewing decisions.

There is therefore a need for a centralized system capable of integrating movie information into a single analytical environment.

3.3 Existing System Analysis

Several movie information systems currently exist, including:

- IMDb
- Rotten Tomatoes
- Letterboxd
- Netflix
- Prime Video
- Disney+

These platforms provide valuable information but suffer from several limitations.

Limitations of Existing Systems

- Lack of cross-platform rating comparison.
- Limited visual analytics.
- No hidden gem discovery mechanisms.
- Poor support for genre intelligence.
- Limited streaming platform analysis.
- Absence of integrated dashboards.

Most systems focus on information presentation rather than information analysis.

3.4 Proposed System

To overcome these limitations, Watch Craft proposes a movie analytics dashboard capable of integrating multiple movie metrics and visualizing them through interactive dashboards.

The proposed system provides:

- Multi-platform rating analysis.
- Audience versus critic comparison.
- Genre intelligence.
- Streaming analytics.
- Hidden gem discovery.
- Interactive filtering.
- Movie recommendations.

Unlike traditional movie websites, Watch Craft focuses on analytical exploration rather than simple information retrieval.

3.5 Objectives of the Proposed System

The major objectives of Watch Craft are:

- To create an interactive movie analytics dashboard.
- To compare multiple movie rating systems.
- To identify hidden gems within the dataset.

- To analyze genre trends and distributions.
- To study streaming platform behavior.
- To provide recommendation support.
- To improve movie discovery through visualization.

These objectives guided the development of the platform.

3.6 Requirement Analysis

Requirement analysis helps identify the resources and functionalities necessary for successful implementation.

Functional Requirements

The system should:

- Load movie datasets.
- Perform preprocessing operations.
- Generate visualizations.
- Support dynamic filtering.
- Compare ratings across platforms.
- Recommend movies.
- Display analytical insights.

Non-Functional Requirements

The system should provide:

- Fast response times.
- User-friendly navigation.

- Interactive visualizations.
- High reliability.
- Scalability for larger datasets.
- Attractive user interface.

3.7 Feasibility Study

A feasibility study was conducted to determine whether the proposed system could be developed successfully.

Technical Feasibility

The project is technically feasible because it uses mature and widely supported technologies such as:

- Python
- Streamlit
- Pandas
- Plotly
- NumPy

These technologies provide all the functionality required for dashboard development and data analysis.

Economic Feasibility

The project is economically feasible because all technologies used are open-source and freely available.

The project requires:

- No software licensing costs.
- No database server costs.
- No expensive infrastructure.

This significantly reduces development expenses.

Operational Feasibility

The dashboard was designed for both technical and non-technical users.

Features such as:

- Sidebar navigation
- Interactive filters
- Visual charts
- Responsive design

make the system easy to operate without prior technical knowledge.

Schedule Feasibility

The project was completed successfully within the six-week industrial training period.

Major phases included:

Activity:	Duration:
Dataset Collection	2 Days
Data Cleaning	3 Days
Dashboard Development	3 Days
Visualization Development	4 Days
Testing	2 Days
Documentation	2 Days

The project schedule was realistic and achievable.

3.8 Hardware Requirements

The minimum hardware requirements for running Watch Craft are:

Component:	Requirement:
Processor	Intel Core i3 or above
RAM	4 GB
Storage	500 MB Free Space
Display	1366 × 768 Resolution

Recommended configuration:

Component:	Recommended:
Processor	Intel Core i5 or above
RAM	8 GB
Storage	SSD Storage
Display	Full HD Resolution

3.9 Software Requirements

The software environment required for the project is:

Software	Purpose
Python	Core Programming Language
Streamlit	Dashboard Development
Pandas	Data Processing
NumPy	Numerical Operations
Plotly	Interactive Visualization
Matplotlib	Statistical Visualization
VS Code	Development Environment

3.10 Target Users

The platform is intended for:

Movie Enthusiasts

Users interested in discovering movies and trends.

Researchers

Researchers studying entertainment data and audience behavior.

Students

Students learning data science and dashboard development.

Analysts

Professionals interested in entertainment analytics.

3.11 Advantages of the Proposed System

The proposed system offers several advantages over traditional movie websites:

- Centralized analytics platform.
- Interactive visualizations.
- Hidden gem discovery.
- Genre intelligence.
- Streaming analytics.
- Dynamic filtering.
- Better user experience.

These advantages significantly improve movie exploration.

3.12 Summary

This chapter analyzed the problems associated with existing movie information systems and established the need for an integrated analytical platform.

The feasibility study confirmed that Watch Craft is technically, economically, operationally, and schedule-wise feasible. The requirements identified during this stage formed the foundation for the development methodology discussed in the next chapter.

CHAPTER 4: METHODOLOGY

4.1 Introduction

Methodology refers to the systematic approach adopted for the development of a software project. It defines the sequence of activities involved in transforming project requirements into a fully functional system.

Since Watch Craft was developed incrementally with continuous improvements in visualizations, recommendations, and analytical modules, an Iterative Development Model was selected for the project.

This approach allowed features to be developed, tested, improved, and integrated gradually, making the system more flexible and easier to maintain.

The overall development process consisted of the following phases:

1. Dataset Collection
2. Data Preprocessing
3. Feature Engineering
4. Exploratory Data Analysis
5. Visualization Development
6. Dashboard Integration
7. Testing and Validation
8. Deployment

4.2 Development Methodology

Iterative Development Model

The Iterative Development Model divides the project into smaller development cycles where each cycle adds new functionality and improves previous implementations.

Unlike traditional waterfall models, the iterative model supports continuous improvement and feedback incorporation.

The methodology was selected because Watch Craft evolved through multiple stages, including:

- Initial dataset exploration
- Visualization development
- Hidden gem implementation
- Recommendation module integration
- User interface enhancements
- Deployment optimization

Advantages of Iterative Development

- Supports gradual feature addition.
- Simplifies debugging and testing.
- Reduces development risk.
- Allows continuous improvement.
- Improves software quality.

4.3 Project Workflow

The complete workflow of Watch Craft is illustrated below:

Movie.csv Dataset

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Data Loading

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Data Preprocessing

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Feature Engineering

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Exploratory Data Analysis

↓

Visualization Generation

↓

Dashboard Development

↓

Testing

↓

Deployment

Each stage contributes towards transforming raw movie data into meaningful analytical insights.

4.4 Dataset Collection

The first stage involved collecting movie-related information in CSV format.

The dataset contains information such as:

- Movie title
- Release year
- Genre
- Director
- IMDb ratings
- Rotten Tomatoes scores
- Audience ratings
- Streaming platforms
- Language information

The dataset was selected because it provides multiple rating systems and rich metadata suitable for entertainment analytics.

The final dataset used in the project was stored in:

Movie.csv

which serves as the primary data source for all analytical modules.

4.5 Data Preprocessing

Real-world datasets often contain missing values, inconsistent formatting, duplicate records, and incorrect data types.

Therefore, preprocessing was performed before analysis.

The preprocessing operations implemented in Watch Craft include:

Missing Value Handling

Missing values were identified and handled using suitable replacement techniques or record removal methods.

Duplicate Removal

Duplicate movie records were removed to improve analytical accuracy.

Data Type Conversion

Columns were converted into appropriate formats such as:

- Integer
- Float
- String

Data Standardization

Genre names and platform names were standardized to ensure consistency throughout the dataset.

These preprocessing operations improve both visualization quality and computational accuracy.

4.6 Feature Engineering

Feature engineering refers to the process of creating additional attributes from existing data to improve analytical capabilities.

Several derived features were generated in Watch Craft.

Audience-Critic Gap

The difference between audience ratings and critic ratings was calculated using:

Audience-Critic Gap = Audience Rating − Critic Rating

This metric helps identify movies where audience opinions differ significantly from professional reviews.

Hidden Gem Score

The Hidden Gem Detector uses multiple rating systems to identify highly rated but less popular movies.

Factors considered include:

- IMDb Score
- Audience Rating
- Critic Score
- Popularity Indicators

Genre Expansion

Movies belonging to multiple genres were separated into individual genre categories for more accurate analysis.

Example:

Action, Adventure, Science Fiction

was transformed into:

- Action
- Adventure

- Science Fiction

Streaming Platform Extraction

Movies available on multiple platforms were separated into individual platform records for streaming analytics.

4.7 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify important patterns.

The analysis focused on:

- Rating distributions
- Genre popularity
- Audience behavior
- Critic opinions
- Platform distributions
- Correlation among ratings

EDA helped determine which visualizations would provide the most useful insights.

4.8 Visualization Development

Visualization is the core component of Watch Craft.

Interactive visualizations were developed using Plotly and integrated with Streamlit.

The visualizations include:

- Bar Charts
- Scatter Plots
- Histograms
- Pie Charts
- Heatmaps
- Treemaps
- Correlation Matrices

These visualizations enable users to explore movie data interactively.

4.9 Dashboard Integration

After developing individual analytical modules, they were integrated into a unified Streamlit dashboard.

The final application consists of the following modules:

- Explore
- Dataset
- Preprocessing
- Visualization
- Hidden Gems
- Recommendation System
- About

Navigation between modules is managed through the sidebar menu implemented using Streamlit Option Menu.

The dashboard updates dynamically whenever the user modifies filters or selections.

4.10 Testing and Validation

Testing was performed continuously throughout development.

The major testing activities included:

- Dataset validation
- Visualization testing
- Filter testing
- Recommendation testing
- User interface testing

Testing ensured that all modules worked correctly and produced reliable outputs.

4.11 Deployment

After successful testing, the application was deployed using Streamlit Cloud.

The deployed application is available at:

Watch-Craft

Deployment provides:

- Public accessibility

- Cross-device compatibility
- Easy sharing
- Cloud-based hosting

4.12 Summary

This chapter described the methodology adopted during the development of Watch Craft. The Iterative Development Model enabled gradual feature addition and continuous improvement throughout the project lifecycle.

The methodology included dataset collection, preprocessing, feature engineering, exploratory analysis, visualization development, dashboard integration, testing, and deployment.

These stages collectively transformed raw movie data into an interactive movie analytics platform capable of generating meaningful insights and improving movie discovery.

CHAPTER 5: TECHNOLOGIES USED

5.1 Introduction

The successful development of any software project depends heavily on the selection of suitable technologies and tools. Since Watch Craft is a movie analytics and visualization platform, it required technologies capable of handling data processing, statistical analysis, interactive visualizations, and web application development.

The technologies used in this project were selected based on the following criteria:

- Ease of implementation
- Performance
- Scalability
- Open-source availability
- Community support
- Compatibility with data science workflows

This chapter discusses the technologies and software tools used during the development of Watch Craft.

5.2 Python

Python is a high-level, interpreted programming language widely used in Data Science, Artificial Intelligence, Machine Learning, and Web Development.

Due to its simple syntax and extensive ecosystem of libraries, Python has become one of the most popular languages for analytical applications.

Role in Watch Craft

Python served as the core programming language and was used for:

- Data preprocessing
- Feature engineering
- Statistical analysis
- Visualization generation
- Recommendation logic
- Dashboard integration

Advantages

- Simple syntax
- Large library ecosystem
- Cross-platform support
- Rapid development capabilities

5.3 Streamlit

Streamlit is an open-source Python framework designed specifically for creating data science and machine learning web applications.

It allows developers to convert Python scripts into fully functional web applications with minimal effort.

Role in Watch Craft

Streamlit was used for:

- Dashboard development
- User interface design

- Interactive filters
- Sidebar navigation
- Dynamic content updates

The entire Watch Craft platform is built using Streamlit components and widgets.

Advantages

- Fast development
- Interactive widgets
- Real-time updates
- Easy deployment

5.4 Pandas

Pandas is one of the most widely used libraries for data manipulation and analysis in Python.

It provides powerful data structures such as:

- Series
- DataFrames

which simplify data handling operations.

Role in Watch Craft

Pandas was used for:

- Loading Movie.csv
- Data cleaning

- Missing value handling
- Filtering operations
- Aggregation
- Statistical analysis

Advantages

- Efficient data manipulation
- Easy syntax
- Excellent CSV support
- Integration with visualization libraries

5.5 NumPy

NumPy (Numerical Python) is a library used for numerical computations and mathematical operations.

It provides support for multidimensional arrays and optimized numerical functions.

Role in Watch Craft

NumPy was used for:

- Mathematical calculations
- Statistical computations
- Feature engineering
- Numerical transformations

Advantages

- Faster computations
- Efficient memory management
- Optimized mathematical operations

5.6 Plotly

Plotly is an interactive visualization library that enables developers to create modern web-based graphs and dashboards.

Unlike traditional plotting libraries, Plotly provides built-in interactivity.

Role in Watch Craft

Plotly was used to generate:

- Bar Charts
- Scatter Plots
- Histograms
- Heatmaps
- Treemaps
- Pie Charts
- Correlation Graphs

Advantages

- Interactive visualizations
- Hover information
- Zooming and panning
- Responsive design

Most visualizations within Watch Craft were developed using Plotly.

5.7 Matplotlib

Matplotlib is a visualization library used for creating static plots and graphs.

It provides extensive customization options and high-quality graphical output.

Role in Watch Craft

Matplotlib was used for:

- Statistical plotting
- Exploratory Data Analysis
- Supporting visualizations

Advantages

- Flexible customization
- Publication-quality graphs
- Wide community support

5.8 Streamlit Option Menu

The Streamlit Option Menu library provides modern sidebar navigation for Streamlit applications.

It improves user experience by creating professional menu layouts.

Role in Watch Craft

The library was used to create:

- Sidebar menus
- Module navigation
- Page switching functionality

The following modules were implemented using the navigation menu:

- Explore
- Dataset
- Preprocessing
- Visualization
- Hidden Gems
- About

Advantages

- Professional appearance
- Improved navigation
- Better organization

5.9 Base64 Encoding

Base64 encoding is used to convert images into text strings that can be embedded directly into web pages.

Role in Watch Craft

Base64 was used for:

- Background images
- Banner rendering

- Custom styling
- User interface enhancements

This allowed Watch Craft to achieve its cinematic visual appearance.

5.10 Visual Studio Code

Visual Studio Code (VS Code) served as the primary development environment during the project.

Features

- Syntax highlighting
- Debugging tools
- Extension support
- Git integration

Role in Watch Craft

VS Code was used for:

- Writing code
- Debugging
- Testing
- Project management

5.11 Technology Stack

The complete technology stack of Watch Craft is shown below:

Layer:	Technology:
Frontend	Streamlit
Backend	Python
Data Processing	Pandas, NumPy
Visualization	Plotly, Matplotlib
Navigation	Streamlit Option Menu
Dataset Storage	CSV
Deployment	Streamlit Cloud

5.12 Advantages of Selected Technologies

The selected technologies provided several benefits:

- Reduced development cost due to open-source software.
- Fast dashboard development.
- Interactive visualizations.
- Easy deployment.
- High scalability and maintainability.

The combination of these technologies enabled the successful implementation of Watch Craft.

5.13 Summary

This chapter discussed the technologies and tools used during the development of Watch Craft. Python served as the core programming language, while Streamlit provided the framework for dashboard development.

Pandas and NumPy were used for data processing and analysis, whereas Plotly and Matplotlib enabled interactive visualization and exploratory analysis.

The selected technology stack provided an efficient, scalable, and cost-effective solution for building an interactive movie analytics platform.

CHAPTER 6: DATASET DESCRIPTION

6.1 Introduction

The dataset serves as the foundation of the Watch Craft platform. All visualizations, insights, recommendations, and analytics are generated using information stored in the movie dataset.

The project uses a CSV dataset named Movie.csv, containing movie metadata collected from multiple review platforms and streaming services.

6.2 Dataset Overview

Property:	Value:
Dataset Name	Movie.csv
Format	CSV
Domain	Entertainment Analytics
Movies	449
Features	26
Year Range	1920–2025

The dataset contains both numerical and categorical attributes suitable for visualization and analysis.

6.3 Major Attributes

Attribute:	Description:
Title:	Movie name
Year:	Release year
Genre:	Movie genres
Director:	Director name
IMDb_100:	IMDb score
Audience Rating:	Audience score
Critic Rating:	Rotten Tomatoes critics score
Letterboxd:	Letterboxd rating
Streaming On:	Streaming availability

6.4 Dataset Processing

The following preprocessing operations were performed:

- Missing value handling
- Duplicate removal
- Genre standardization
- Data type conversion
- Multi-genre expansion

These operations improved analytical accuracy.

6.5 Importance of Dataset

The dataset powers:

- Rating Explorer
- Genre Intelligence
- Hidden Gem Detector
- Recommendation System
- Streaming Analytics

6.6 Summary

The Movie.csv dataset provides rich movie metadata and multiple rating systems, making it highly suitable for entertainment analytics and visualization.

CHAPTER 7: SYSTEM DESIGN

7.1 Introduction

Watch Craft follows a layered architecture that transforms raw movie data into interactive visual insights.

7.2 Architecture

Movie.csv

↓

Preprocessing

↓

Feature Engineering

↓

Visualization Engine

↓

Streamlit Dashboard

↓

User

7.3 Major Modules

The system consists of:

- Explore Module
- Dataset Module

- Preprocessing Module
- Visualization Module
- Hidden Gems Module
- Recommendation Module
- About Module

7.4 Input Design

Inputs are provided through:

- Dropdown menus
- Sliders
- Sidebar filters
- Search boxes

7.5 Output Design

Outputs include:

- Interactive charts
- Statistical summaries
- Recommendation lists
- Visual dashboards

7.6 Summary

The modular architecture ensures maintainability, scalability, and smooth user interaction.

CHAPTER 8: FEATURES OF WATCH CRAFT

The major features implemented in Watch Craft include:

Explore Module:

Provides project introduction and navigation.

Dataset Explorer:

Displays dataset information and statistics.

Preprocessing Module:

Shows cleaning and transformation steps.

Rating Explorer:

Compares ratings from multiple platforms.

Genre Intelligence:

Analyzes genre trends and popularity.

Streaming Analytics:

Studies content distribution across platforms.

Hidden Gem Detector:

Identifies highly rated but underrated movies.

Recommendation System (Currently Under development):

Suggests movies based on filters and similarity.

Interactive Visualizations:

Includes:

- Bar charts
- Scatter plots
- Heatmaps
- Treemaps
- Histograms

Dark Cinematic Interface:

Provides an immersive user experience.

These features collectively transform traditional movie exploration into a data-driven analytical experience.

CHAPTER 9: SCREENSHOTS AND IMPLEMENTATION

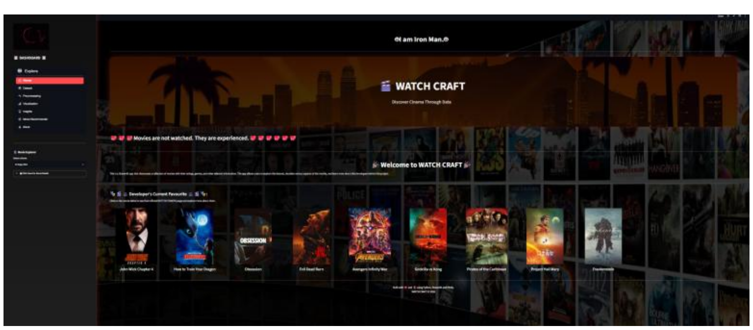
The implementation of Watch Craft consists of multiple dashboard modules.

The following screenshots should be included:

Figure No.:	Screenshot:
Figure 9.1	Explore Page
Figure 9.2	Dataset Module
Figure 9.3	Preprocessing Module
Figure 9.4	Visualization Dashboard
Figure 9.5	Genre Intelligence
Figure 9.6	Hidden Gems
Figure 9.7	Recommendation Module
Figure 9.8	About Page

Each screenshot demonstrates successful implementation of the corresponding functionality.

The deployed application confirms successful integration of all modules into a unified dashboard environment.

Figure 9.1: 

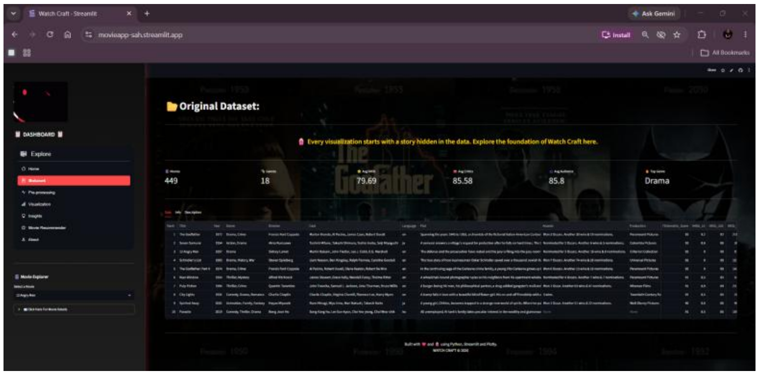


Figure 9.2:

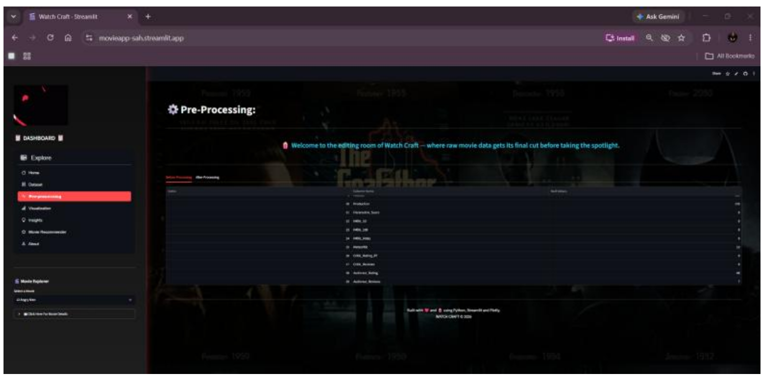


Figure 9.3:

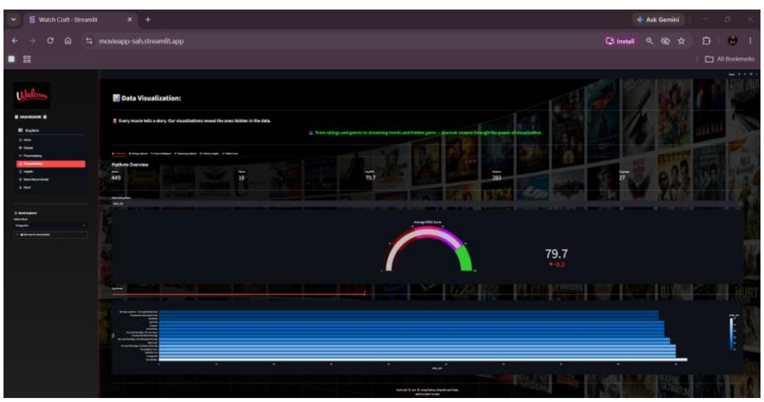


Figure 9.4:

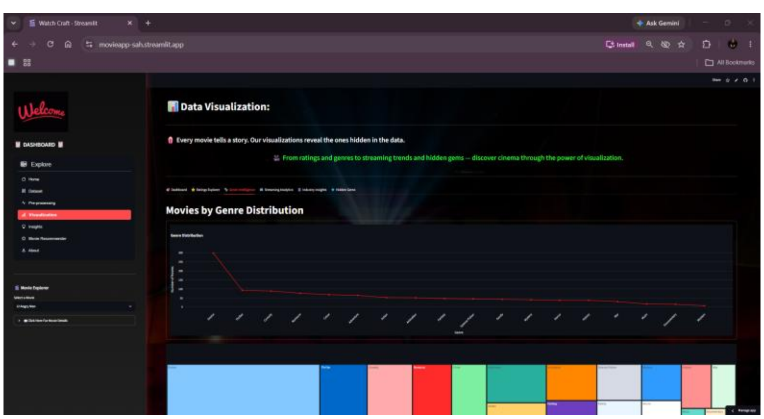


Figure 9.5:

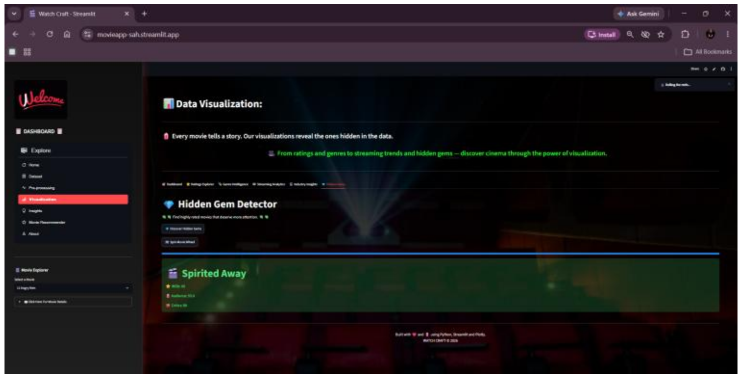
Figure 9.6: 

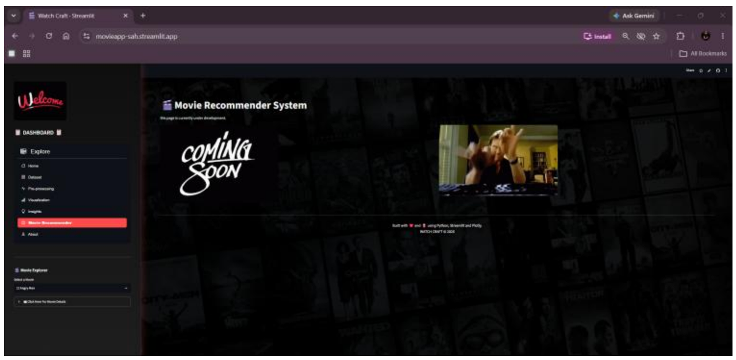
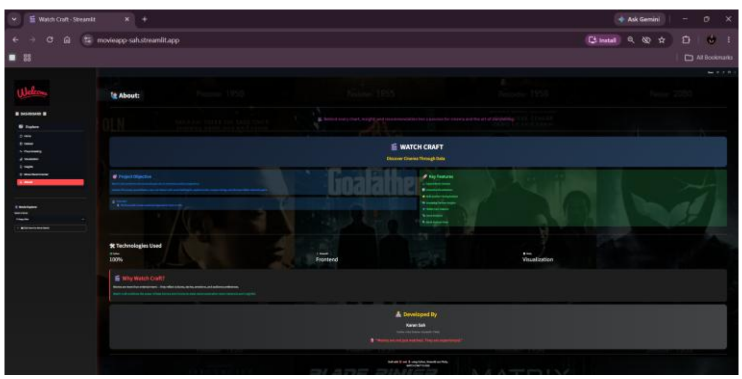
Figure 9.7: 

Figure 9.8: 

CHAPTER 10: TESTING AND VALIDATION

Testing was performed to ensure system reliability and correctness.

Unit Testing:

The following components were tested:

- Dataset loading
- Missing value handling
- Filtering functions
- Recommendation logic

Integration Testing:

Verified interaction between:

- Dataset and visualizations
- Filters and dashboards
- Recommendations and analytics

Performance Testing:

Metric:	Result:
Dataset Loading	<2 seconds
Visualization Rendering	<3 seconds
Navigation	Smooth

User Acceptance Testing:

Users reported:

- Easy navigation
- Attractive interface
- Useful recommendations
- Fast response times

The successful completion of testing confirms the reliability of Watch Craft.

CHAPTER 11: RESULTS AND DISCUSSION

The analysis performed by Watch Craft generated several meaningful insights.

Major Findings:

- Audience and critics often disagree.
- Drama dominates the dataset.
- Certain streaming platforms specialize in specific genres.
- Highly rated movies are not always the most popular.

Hidden Gems:

The Hidden Gem Detector successfully identified underrated movies with strong ratings across multiple platforms.

Genre Trends:

Drama and Thriller showed the highest representation within the dataset.

Rating Correlation:

IMDb and Audience Ratings exhibited strong positive correlation.

The results demonstrate the effectiveness of interactive visual analytics in entertainment datasets.

CHAPTER 12: FUTURE SCOPE

Future enhancements for Watch Craft include:

- AI recommendation systems
- Sentiment analysis of reviews
- Real-time API integration
- User accounts and personalization
- Watchlists
- Mobile application development
- NLP-based plot analysis
- Generative AI movie assistant

These improvements can transform Watch Craft into a complete entertainment intelligence platform.

CHAPTER 13: CONCLUSION

Watch Craft successfully demonstrates the application of Data Science and Visualization in the entertainment industry.

The project integrates:

- Multi-platform ratings
- Genre intelligence
- Streaming analytics
- Hidden gem discovery
- Recommendation support

The platform transforms raw movie data into meaningful visual insights while providing users with an engaging analytical experience.

The successful deployment of Watch Craft establishes a strong foundation for future AI-driven entertainment analytics systems.

CHAPTER 14: REFERENCES

Documentation:

- Python Documentation
- Streamlit Documentation
- Plotly Documentation
- Pandas Documentation

Data Sources:

- IMDb
- Rotten Tomatoes
- Letterboxd

Project Deployment:

- Watch Craft Live Application